4. For the ideal transformer circuit shown in Figure 11, determine the turn ratio n so that the 24Ω resistor receives maximum average power, and find this maximum power. **(15 marks)**

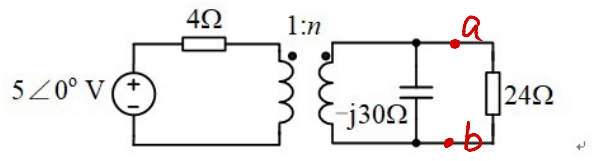
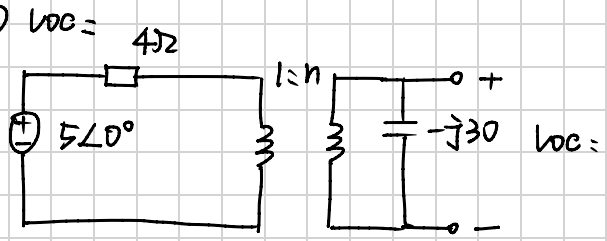


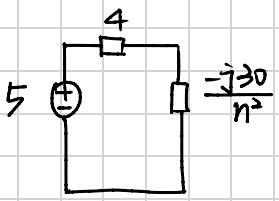
Figure 11

求ab两端的戴维南电路

① $V_{oc}=$



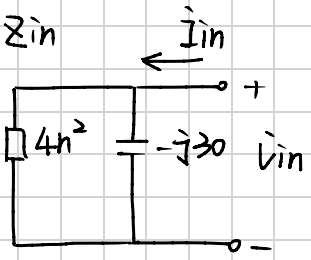
$V_{oc}=$



$$\dot{V}_1 = 5\cdot\frac{\frac{-j30}{n^2}}{4-\frac{j30}{n^2}}$$

$$V_{oc} = n\dot{V}_1 = \frac{\frac{-j150}{n}}{4-\frac{j30}{n^2}}$$

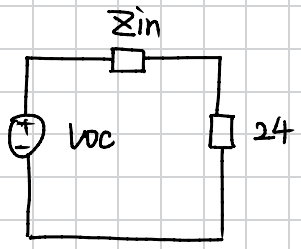
② Z_{in}



$$Z_{in} = \frac{-j120n^2}{4n^2-j30} = \frac{-j120n^2(4n^2+j30)}{(4n^2)^2+(30)^2}$$

$$= \frac{3600n^2}{(4n^2)^2+30^2} - \frac{j4800n^4}{(4n^2)^2+30^2}$$

③ 等效电路：



$$|Z_{in}| = \frac{120n^2}{\sqrt{(4n^2)^2+30^2}} = 24$$

$$n=\sqrt{10}$$

$$V_{oc} = \frac{\frac{-j150}{\sqrt{10}}}{4-j3} \qquad |V_{oc}| = \frac{\frac{150}{\sqrt{10}}}{5} = \frac{30}{\sqrt{10}} = 3\sqrt{10}$$

$$n=\sqrt{10}$$

$$Z_{in} = \frac{36000}{1600+900} - \frac{j48000}{1600+900}$$

$$= \frac{36000}{2500} - \frac{j48000}{2500}$$

$$= 14.4 - j19.2$$

$$\frac{\dot{V}_{oc}}{38.4-j19.2} = I_s, \quad \dot{V}_{oc} = \frac{\frac{-j150}{\sqrt{10}}}{4-j3}$$

$$I_s = \frac{\sqrt{10}}{16} - j\frac{\sqrt{10}}{32}$$

$$P = |I_s|^2 \cdot R = \left(\frac{5\sqrt{2}}{32}\right)^2 \cdot 24$$

$$= 1.1718\,W$$